BASE CHANGE OF THE SECTIONS WITH SUPPORT ALONG A FLAT RING MAP, AND THE BOUNDED CASE WHEN THE RINGS HAVE FINITE DIMENSION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The description of the local cohomology with support in a specialization-stable subset as the filtered colimit of the local cohomologies with support in the closed subsets contained in it. Author lemma lemma-adjoint-ext, whose statement is reproduced verbatim as a context span of this card (source lines 996-1004). Reduces the general support to the closed case by passing to the colimit over the closed subsets, on both sides of the base change map.
- The base change statement for the sections with support in a closed subset along a flat map, in the form recorded in the chapter on dualizing complexes. Invoked inside the counted proof at source lines 1118-1119 as a cross-chapter pointer. Provides the closed case from which the general case is assembled.

Source reference. Original reference dualizing-lemma-torsion-change-rings: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 741–741. Navigation pointer only: the counted text refers to the Stacks item dualizing-lemma-torsion-change-rings, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $A \to B$ be a flat homomorphism of Noetherian rings. Let $T \subset \mathrm{Spec}(A)$ be a subset stable under specialization. Let $T' \subset \mathrm{Spec}(B)$ be the inverse image of T. Then the canonical map*

$$R\Gamma_T(K) \otimes^{\mathbf{L}}_A B \longrightarrow R\Gamma_{T'}(K \otimes^{\mathbf{L}}_A B)$$

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is an isomorphism for $K \in D^+(A)$. If A and B have finite dimension, then this is true for $K \in D(A)$.

Lemma 0.2. *Let A be a Noetherian ring. Let $T \subset \mathrm{Spec}(A)$ be a subset stable under specialization. For any object K of $D(A)$ we have*

$$H^i(RH^0_T(K)) = \mathrm{colim}_{Z \subset T \text{ closed}} H^i_Z(K)$$

Proof. From the map $R\Gamma_T(K) \to K$ we get a map $R\Gamma_T(K) \otimes^{\mathbf{L}}_A B \to K \otimes^{\mathbf{L}}_A B$. The cohomology modules of $R\Gamma_T(K) \otimes^{\mathbf{L}}_A B$ are supported on T' and hence we get the arrow of the lemma. This arrow is an isomorphism if T is a closed subset of $\mathrm{Spec}(A)$ by Dualizing Complexes, Lemma dualizing-lemma-torsion-change-rings. Recall that $H^i_T(K)$ is the colimit of $H^i_Z(K)$ where Z runs over the (directed set of) closed subsets of T, see Lemma 0.2. Correspondingly $H^i_{T'}(K \otimes^{\mathbf{L}}_A B) = \mathrm{colim}\, H^i_{Z'}(K \otimes^{\mathbf{L}}_A B)$ where Z' is the inverse image of Z. Thus the result because $\otimes_A B$ commutes with filtered colimits and there are no higher Tors. $\square$